Global EV Market Intelligence Dashboard

Project Report | Date: 2 November 2026 | Team ID: [add Team ID]

1. Introduction

1.1 Project Overview

This project builds an interactive Tableau dashboard that turns a large electric vehicle (EV) market dataset (2018 to 2026) into clear answers for three business users. The source data came as two files, ev_market_train.csv (200,000 rows) and ev_market_test.csv (100,000 rows), each with 37 columns. After merging them and removing 1 duplicate row, 299,999 records remained, covering 200 manufacturers, 15 vehicle categories, 5 battery types, 10 manufacturing countries and 7 market regions.

The finished dashboard has four KPI cards, nine visualizations (V1 to V9) and three filters (Model Year, Category and Battery Type). It was published to Tableau Public. Snowflake was not used; the data was prepared with Python and CSV/Excel files and analysed in Tableau Desktop.

1.2 Purpose

The purpose is to give EV market decision makers one place to see sales, leading manufacturers, battery technology adoption, driving range, customer satisfaction and manufacturing geography, instead of working through 300,000 raw records. The dashboard is designed for three personas:

- **Emily Johnson**, Market Strategy Director: sales by vehicle category, leading manufacturers, satisfaction by segment, battery technology adoption and range by category.
- **David Chen**, BI Manager: manufacturer market share, annual sales, battery technology distribution, manufacturing country analysis and customer ratings.
- **Dr. Sarah Williams**, Sustainable Mobility Researcher: comparison of battery technologies, range and capacity, global manufacturing distribution and satisfaction by EV category.

2. Ideation Phase

2.1 Problem Statement

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Emily Johnson, Market Strategy Director at an EV company	decide which vehicle categories, manufacturers and battery technologies to focus on	the market data covers about 300,000 records and no single view links sales, range and ratings	the data is spread across many categories, regions and files and has to be combined by hand	overwhelmed and unsure about strategy decisions
PS-2	David Chen, BI Manager	report manufacturer market share, annual sales and battery technology distribution to leadership	every report needs manual aggregation and there are no ready-made KPIs or filters	the raw data has no consolidated, interactive reporting layer	frustrated and short of time

2.2 Empathy Map Canvas

Persona: **Emily Johnson (Market Strategy Director)**

Says	Thinks
Which vehicle categories are growing the fastest? Who are the leading manufacturers? Show me customer satisfaction for each segment.	Where should we invest next, and in which category and battery technology? I have too many reports and no single view. Can I trust these numbers enough to present to leadership?

Does	Feels
Reviews sales, range and rating data across categories and regions. Compares manufacturers and battery types. Presents findings and recommendations to leadership.	Overwhelmed by about 300,000 records spread across many files. Under pressure to make quick, confident decisions. Confident when clear, visual evidence is available.

2.3 Brainstorming

Group	Idea	Category	Feasibility	Importance	Selected
1	Sales by category bar chart, yearly sales trend and KPI cards	Market Performance	High	High	Yes
2	Top 10 manufacturers, market-share treemap, battery share donut and rating heat map	Competitive Analysis and Battery Technology	High	High	Yes
3	Manufacturing-country map, capacity vs range scatter plot and average range by category	Sustainability and Range	Medium	Medium	Yes

3. Requirement Analysis

3.1 Customer Journey Map

The journey below follows Emily Johnson from raw data to a decision.

Stage	What the user does	Pain point	How the dashboard helps
Need	Gets a request to recommend where to focus in the EV market	Unclear where to start	KPI cards give the headline numbers at once
Explore	Looks at sales by category, manufacturer and battery type	Data is spread over large files	Nine linked visuals on one page

Stage	What the user does	Pain point	How the dashboard helps
Narrow down	Filters by year, category or battery type	Manual re-cutting of data for each question	Model Year, Category and Battery Type filters, plus click-to-filter on V1, V4 and V7
Insight	Compares segments, ranges and ratings	Hard to see which differences matter	Charts show where gaps are large (category, battery type) and where they are tiny
Decide and report	Shares findings with leadership	Needs a shareable, trusted view	Published Tableau Public dashboard with verified totals

3.2 Solution Requirements

Functional requirements

1. Combine the train and test files and remove duplicates.
2. Show total sales, average rating, average range and average battery capacity as KPIs.
3. Provide nine visuals: sales by category, top 10 manufacturers, market share, battery type share, average range by category, rating heat map, sales by manufacturing country, capacity vs range, and sales trend by model year.
4. Provide filters for Model Year, Category and Battery Type, and click-to-filter on V1, V4 and V7.
5. Publish the dashboard so it can be shared by link.

Non-functional requirements

- Totals in the dashboard must match the raw data.
- The dashboard must stay readable at a fixed size of 1400 x 1700.
- The prepared data must be small enough to load quickly (an aggregated extract of 45,478 rows plus 200 manufacturer rows).

3.3 Data Flow Diagram

1. **Source:** ev_market_train.csv (200,000 rows) and ev_market_test.csv (100,000 rows).

2. **Preparation (Python, pandas):** merge, check missing values, remove duplicates, trim text, validate fields.
3. **Combined file:** ev_market_combined.csv (299,999 rows).
4. **Aggregation:** EV_Agg (45,478 rows, one row per model year, category, battery type, manufacturing country and market region) and Manufacturer (200 rows), saved in EV_Tableau_Extract.xlsx.
5. **Tableau Desktop:** connect to the Excel file, create calculated fields (Avg Range, Avg Rating, Avg Capacity, YoY Growth %), build sheets and the dashboard.
6. **Tableau Public:** extracts are created and the dashboard is published and shared by link.

3.4 Technology Stack

Layer	Tool
Data preparation	Python (pandas)
Prepared data	CSV and Excel (EV_Tableau_Extract.xlsx)
Analysis and dashboard	Tableau Desktop
Publishing	Tableau Public (with data extracts)

4. Project Design

4.1 Problem Solution Fit

Problem	Solution in this project
Too many records and no single view	One dashboard with KPIs and nine linked visuals
Manual aggregation for every report	Pre-aggregated EV_Agg and Manufacturer tables with additive columns
No quick way to slice the data	Three filters and click-to-filter on V1, V4 and V7
Need for a trustworthy, shareable result	Totals checked against raw data and dashboard published on Tableau Public

4.2 Proposed Solution

Visual	What it shows	Main persona served
V1 Sales by Category	Total sales for each of 15 categories	Emily
V2 Top 10 Manufacturers	Leading manufacturers by sales	Emily, David
V3 Market Share	Treemap of manufacturer share	David
V4 Battery Share	Share of sales by battery type	Emily, David, Sarah
V5 Avg Range by Category	Average range for each category	Emily, Sarah
V6 Rating Heat Map	Average rating by category and battery type	David, Sarah
V7 Country Map	Sales by manufacturing country	David, Sarah
V8 Capacity vs Range	Battery capacity against range, with a trend line	Sarah
V9 Sales Trend	Sales by model year for categories	Emily, David

4.3 Solution Architecture

The solution has four layers: **raw data** (two CSV files), **prepared data** (the combined CSV and the Excel extract with EV_Agg and Manufacturer), **analysis** (Tableau Desktop with calculated fields) and **delivery** (the dashboard published on Tableau Public). V2 and V3 use the Manufacturer table and the other visuals use EV_Agg, so the dashboard filters affect only the EV_Agg visuals.

5. Project Planning and Scheduling

5.1 Project Planning

The work followed these phases in order. [Add start and end dates and team member roles.]

Phase	Main activities
Ideation	Problem statements, empathy map, brainstorming and prioritization
Requirement analysis	Customer journey, requirements, data flow and technology stack
Project design	Problem-solution fit, proposed solution and architecture
Data preparation	Null check, duplicate removal, cleaning, aggregation and validation
Dashboard development	Calculated fields, nine visuals, KPI cards, filters and layout
Testing	Total checks against raw data, KPI and filter checks
Publishing and documentation	Tableau Public publishing, report and demonstration

6. Functional and Performance Testing

6.1 Performance Testing

Test	Result
Total sales in the extract against raw data	Matches: 2,323,303,050 units
Average rating in the extract against raw data	Matches: 4.0874 (shown as 4.09)
Row counts after preparation	299,999 combined rows, 45,478 in EV_Agg and 200 in Manufacturer
KPI cards	Show 2,323,303,050 sales, 4.09 rating, 324 km range and 78 kWh battery
Filters (Model Year, Category, Battery Type)	Apply to every visual that uses EV_Agg
Click-to-filter on V1, V4 and V7	Updates the other EV_Agg visuals

Test	Result
Publishing	Dashboard published to Tableau Public after both data sources were set to extracts

Known limitation: V2 and V3 use a separate data source, so the dashboard filters do not change them. Every individual visual was not checked against the raw data; the totals and KPI values were. [Add load-time or other test results if you measured them.]

7. Results

7.1 Output Screenshots

[Insert screenshots here: (1) full dashboard, (2) KPI row, (3) V1 to V9 individually, (4) a filtered view, (5) the published Tableau Public page.]

Key findings from the data

- Total sales are 2.32 billion units, with an average rating of 4.09, an average range of 324 km and an average battery of 78 kWh.
- The largest categories by sales are Two-Wheeler Scooter (about 260M), Sedan (232M), Hatchback (231M) and Compact SUV (230M). The smallest are Electric Bus (16M), Pickup (19M) and Electric Truck (20M).
- Range depends strongly on category: Luxury Sedan averages 550 km and Luxury SUV 525 km, against 105 km for Two-Wheeler Scooters.
- By share of sales, NMC leads with 33.7%, then LFP 32.0%, NCA 17.3%, Sodium-Ion 9.9% and Solid-State 7.1%.
- Manufacturers, countries and ratings are almost the same: the top manufacturer, AmpereWorks, has only 0.59% share, China leads manufacturing at 237M units against 227M for Japan, and ratings stay between about 4.05 and 4.13. The dataset appears synthetic, so only category and battery type differences are meaningful.

8. Advantages and Disadvantages

Advantages

- One interactive page answers the questions of all three personas.
- Pre-aggregated data keeps the dashboard fast and the totals easy to verify.

- Filters and click-to-filter let users explore without writing queries.
- The result is shareable by link through Tableau Public.

Disadvantages

- The dataset looks synthetic, so most manufacturer, country and rating differences are very small and should not be read as real advantages.
- V2 and V3 use a separate data source and do not respond to the dashboard filters.
- Tableau Public requires extracts and makes the workbook and data public.
- Data preparation was done with Python and files rather than a warehouse such as Snowflake, so it is less automated.

9. Conclusion

The project turned 299,999 EV market records into a single dashboard of four KPIs and nine visuals that serve a strategy director, a BI manager and a researcher. The numbers were checked against the raw data, and the dashboard is published for sharing. The main real signals are the differences between vehicle categories and battery types, while manufacturers, countries and ratings are nearly uniform, which the report states openly.

10. Future Scope

- Replace the file-based preparation with an automated pipeline or data warehouse (for example Snowflake).
- Add live or real-world EV market data to replace or validate the synthetic data.
- Bring the manufacturer data into the same model so that every visual responds to the filters.
- Add forecasting of sales by category and battery type.
- Add price and charging-time analysis, and mobile-friendly layouts.

11. Appendix

Source Code

The full preparation code is in the Data Preparation document. Its main steps are:

```
import pandas as pd
train = pd.read_csv("ev_market_train.csv")
test = pd.read_csv("ev_market_test.csv")
df = pd.concat([train, test], ignore_index=True)
df = df.drop_duplicates()          # 300000 -> 299999 rows
keys = ["model_year", "category", "battery_type",
        "manufacturing_country", "market_region"]
agg = df.groupby(keys).agg(
    models=("vehicle_id", "count"),
    sales_units=("annual_sales_units", "sum"),
    rating_sum=("customer_rating", "sum"),
    rating_n=("customer_rating", "count")).reset_index()
```

Dataset Link

[Add the link to the dataset source.] Files used: ev_market_train.csv, ev_market_test.csv, ev_market_combined.csv and EV_Tableau_Extract.xlsx.

GitHub and Project Demo Link

[Add the Tableau Public dashboard link and any GitHub or Drive link.]